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A Brief Introduction to Clustering in Computer ScienceAmr S. Ghoneim,
Ph.D.(Assistant Professor, Computer Science Dept.) . Los Angeles University of
California Press. I think that this publication will strengthen the visibility of the
European Union in the field of human rights considerably.

What is Cluster Analysis? Finding groups of objects such that the objects in a group will be similar (or related) to one another and different from (or unrelated to) the objects in other groups. Based on information found in the data that describes the objects and their relationships. - Also known as unsupervised classification. Score, plus many applications, haircut haircut Understanding, Understanding, group related documents for browsing, or to find genes and proteins that have similar functionality. Bare minimum makeup with shades of some glow on those sculpted cheeks and tresses tied in a messy ponytail, the actor looked like a princess. In the course of his reconstitution of the urban administration he created new offices in connection with the public works, streets, and aqueducts of Rome.

[illegible]

Notion of a Cluster is Ambiguous in the following areas: 1) Algorithmic Diversity: Different clustering algorithms impose distinct assumptions about cluster structure. For example, 1. K-means assumes spherical, equally sized clusters. 2. DBSCAN identifies dense regions separated by sparse areas. 3. Hierarchical clustering builds nested groupings based on proximity. These varying assumptions lead to divergent results, as each algorithm "sees" clusters through its own lens. 2) Subjectivity in Criteria Clusters can be defined by density, connectivity, centroid proximity, or statistical distributions. Without a ground truth, the choice of criteria is often subjective. 4 - 4 - 4 -

Notion of a Cluster is Ambiguous 3) Data Characteristics: 1. Overlap/Ambiguity: Data may lack clear separation, making non-boundaries between clusters fuzzy (e.g., overlapping Gaussian distributions). 2. High Dimensionality: In high-dimensional spaces, distance and dimensionality are the "metrics lose discriminative power (the "curse of dimensionality")", complicating cluster identification. 3. Noise/Outliers: Some methods treat outliers as noise (e.g., DBSCAN or DBSDSCAN), while others force them into clusters (e.g., k-means), altering interpretations. 4) Parameter Sensitivity: Some algorithms require user-defined parameters to be able to detect outliers in clusters. 5) User-defined Parameters: Many algorithms use user-specified parameters (e.g., number of clusters k , density thresholds). Small changes in these parameters can yield vastly different results, and there is often no objective way to measure them, let alone set them.

Notion of a Cluster is Ambiguous 3) Data Characteristics: 1. Overlap/Ambiguity: Data may lack clear separation, making it difficult to distinguish between clusters or subboundaries. Density between thresholds clusters are fuzzy parameters (e.g., overlapping used in density-based Gaussian distributions). Clustering algorithms (e.g., DBSCAN) to define the minimum conditions required for a region of data points to qualify as a 2. High Dimensionality, 1.5. In high-dimensional spaces, distance clusters. They typically involve two components: 1 metrics lose discriminative power (the "curse of the distance from to ") and •Distance threshold (ϵ): The radius within which a cluster is located ("the $\hat{m}$ dimensionality"), complicating cluster identification. Points are considered part of the same cluster. 3. Noise/Outliers: Some methods treat outliers as noise (e.g., k-means), while others treat them as outliers. •Minimum points (minPts): The number of points required DBS (the minimum points required for a region to be classified as "dense"), while others force them into clusters (e.g., d-mans), using the ϵ -radius to label a region as "Dense." Altering interpretations. 4) Parameter Sensitivity: Many algorithms require user-defined parameters (e.g., number of clusters, density thresholds). Small changes in these parameters can yield vastly different results, and there is often no objective way to measure them, let alone set them.

Notion of a Cluster is Ambiguous. 5) Context-Dependent Goals: The purpose of clustering (e.g., anomaly detection vs. customer segmentation) influences what qualifies as a meaningful cluster. For example, one application might prioritise small, dense clusters, while another seeks balanced groupings. Example 1: Anomaly Detection and Analysis • Goal: Separate "normal" data (clusters) from rare outliers. • Cluster Criteria: Focus on large, dense regions as normal. Treat small/sparse regions of the world as teras anomalies. Definition Parameter Choice to High Minits or Large Minits to ignore small groups. Example 2: Customer Segmentation • Goal: Identify

distinct, interpretable groups for targeted marketing. • Cluster Criteria: Balanced cluster sizes for actionable insights. • Parameter Choice: Predefined k (number of segments) based on business needs.

Notion of a Cluster is Ambiguous and Unlabeled.6) Validation Challenges: Identifying a Cluster Is Ambiguous Without labeled data, evaluating clusters relies on internal metrics that can be used to identify clusters(e.g., Silhouette Score, Davies-Bouldin index), which may conflict across algorithms. This lack of consensus on evaluation exacerbates ambiguity. 8 - 8 -

Types of Clusterings • A clustering is a set of clusters. • • • • One crucial distinction is between non-hierarchical and partitional sets of clusters. Partitional Clustering - A division of data objects into non-overlapping subsets (clusters) such that each data object is in exactly one subset. Hierarchical clustering - A set of nested clusters organised as a hierarchical tree.

Partitional Clustering Preferences, Original Points 1, 2, 3, 4, 5, 6, 7, 8, 9, 11, 12, 13, 14, 15, 16, 17, 18, 20, 19, 21, 22, 25, 29, 26, 28, 30, 34, 39, 35, 38, 45, 44, 46, 39. A Partitional clustering Procedures10.

Hierarchical Clustering, p1, p3, p4, p2, p5, p6, p7, p8, p9, htt httpactory Traditional Archarchical clustering 1, 2, 3, 4, 5, 6, 7, 8, 9, 12, 12 Traditional Dendrogram Graphics, 11.

Types of Clusterings are as follows: Other important distinctions are between: • Exclusive versus Overlapping (non-exclusive) or Fuzzy (probabilistic). • • • • Complete versus Partial. O Exclusive: Each data point belongs to exactly one cluster. E.g., K-means, Hierarchical Clustering. Ance o. Points can belong to multiple clusters (hard assignments). E.g., Overlapping community detection. O Fuzzy (Probabilistic) points have probabilistic memberships (e.g., 0.7 in Cluster A, 0.3 in Cluster B). E.g., Fuzzy C-means, Gaussian Mixture Models. Ance o. Complete, all points are assigned to clusters (no unlabelled data). E.g., K-means, or Non-Agglomerative Clustering. Some points are excluded as noise/outliers. E.g., DBSCAN, HDBSCAN.

Types of Clusterings are: Other important distinctions are between: Why • Â Why This Matters? Versus Overlapping (non-exclusive) versus Fuzzy Application Alignment: (probabilistic). • Use exclusive clustering for customer segmentation (distinct groups). .2.0.0 Complete.0 Use fuzzy versus clustering Partial.1. For text topics (documents can belong to multiple topics). • Exclusive: • Use Each partial data clustering point belongs to an anomaly exactly one detection cluster. (separate E.g., K-means, outliers). Hierarchical Clustering. Ance o. Points can belong to multiple clusters (hard assignments). E.g., Cluster A, Cluster B. Clustering is rarely one-size-fits-all-the choice

depends on data structure, data type, and community size. Overlapping community detection. Cluster A: Cluster B has its own set of data types, subgoals, and interpretability needs! o Fuzzy (Probabilistic): Points have probabilistic memberships (e.g., 0.7 in cluster A, 0.3 in Cluster B). E.g., Fuzzy C-means, Gaussian Mixture Models. Ance o. Complete, all points are assigned to clusters (no unlabelled data). E.g., K-means, or Non-Agglomerative Clustering. 13o Partial: Some points are excluded as noise/outliers. E.g., DBSCAN, HDBSCAN.

Types of Clusters: Well-Separated • Well-Separated Clusters: - A cluster is a set of points such that any point in the cluster is closer (or more similar) to every other point in the cluster than to any point not in the cluster.

Types of Clusters: Center-Based Clusters • Center-based Clusters- A cluster is a set of objects such that an object in a cluster is closer (more similar) to the "center" of a cluster, than to the center of any other cluster. The center of a cluster is often a centroid, the average of all the points in the cluster, or a medoid, the most "representative" point of a cluster.

Types of Clusters: Contiguity-Based • Contiguous Cluster (Nearest neighbor or transitive) - A cluster is a set of points such that a point in a cluster is closer (or more similar) to one or more other points in the cluster than to any point not in the cluster.

Types of Clusters: Density-Based Clusters - A cluster is a region of points, which is separated by low-density regions, from other regions of high density. - Used when the clusters are irregular or intertwined, and when noise and outliers are present. - The three curves don't form clusters since they fade into the noise, as does the bridge between the two small circular clusters.

Similarity and Dissimilarity • Similarity - Numerical measure of how similar two data objects are. Latency is higher when objects are more alike. Itude-2 often falls in the range (0.0,0.1,1.0) perceptual-2.0 Dissimilarity perceptual-4.0 Numerical measure of how different two data objects are. - Is lower when objects are more alike.- Minimum dissimilarity is often 0.5- Upper limit varies between objects. - Upper limit can vary from object to object. • Proximity refers to a similarity or dissimilarity between two objects.19

Euclidean Distance, Ignoring, Chaitin, Distance, Nightmare, Distortion 1, 1, 2, 3, 4, 5, 6, 7, 8, 9, 11, 12, 13, 16, 15, 16 and 16. Itude ($p-q$)², 1, 2, 3, 4, 5, 6, 7, 8, 9, 11, 12, 16, 17, 18, 19, 20, 21, 22, 24, 25, 29, 32, 33, 35, 38, 39, 45, 37, 38 and 39 k, respectively, where n is the number of dimensions (attributes) and pk, and qare, both respectively, the kth attributes (components) or data objects, p and q. Clearly take-off slots at popular times of the day can be critical in

Common Properties of a Distance • Distances, such as the Euclidean distance, have some well-known properties: 1. $D(p, q) \geq 0$ for all p and q and $d(p, q) = 0$ for all p and q . Only if $p = q$. 2. $D(p, q) = d(q, p)$ for all p and q . $D(p, r) \leq d(p, q) + d(q, r)$ for all points p, q , and r . (Triangle Inequality) - where $d(p, q)$ is the distance (dissimilarity) between points (data objects) p and q . A distance that satisfies these properties is a "metric" distance.

Common Properties of a Similarity, such as: • Similarities, also have some well-known properties, including: properties: 1. $s(p, q) = 1$ (or maximum similarity) only if $p = q$. 2. $s(p, q) = s(q, p)$ for all p and q . (Symmetry) - where $s(p, q)$ is the similarity between points (data objects), p and q .

K-means Clustering. • Partitional clustering approach. • • • • Each cluster is associated with a centroid (center point). • • • • Each point is assigned to the cluster with the closest centroid. • Number of clusters, K , must be specified. • • • • The basic algorithm is very simple. The sage Vatsya said this, and so on.

- K-means Clustering Details • Initial centroids are often chosen randomly. - Clusters produced vary from one run to another. • • • • The centroid is (typically) the mean of the points in the center of the subcluster. • 'Closeness' is measured by Euclidean distance, cosine similarity, correlation, etc. • K-means will converge for common similarity measures mentioned above. • Most of the convergence happens in the first few iterations. Itude-1 Often, the stopping condition is changed to 1 until relatively few points change clusters' severity-2. Ance Complexity is $O(n, n, y, y) K.C. I * d$ - n = number of points, K = number of clusters, I = number of iterations, d = number of attributes 24.

- K-means Clustering Details • Initial centroids are often chosen randomly. - Clusters produced vary from one run to another. • • • • The centroid is (typically) the mean of the points in the center of the subcluster. • 'Closeness' is measured by Euclidean distance, cosine distance, and cosine similarity. Suppose: similarity, correlation, etc. -plus net, 10,000,000 points, 1,943,621,1, http.com, 4,937,837, 5,746, 6,739, 619, 819, 919, 719, 11,832, 12,732, 9,632, 8,737, 9.6, 8.7, 11.7. $K = 20$ clusters, not the ones mentioned above. .2x2x0x0.0x2.00.1x3.3.0.2.1.0, Score Most of the convergence happens in the first few iterations. .2x0.0x0x2x1x0,0x3x2.0 Often, the stopping condition is changed to 0 til relatively few points, perceptual Total operations ■s 10,00 0,10,000x1,0000x1.5x2,5x3,065,066,067,068,039,064,069,032,049,037,046,042,036, 043,034,038,041,053,051,061 Itude change clusters' perceptual complexity complexity is $O(n) n, n, y, y.c. K.C. I * d$ - n = number of points, K = number of clusters, I = number of iterations, d = number of attributes 25.

Evaluating K-means Clusters of Clusters • Most common measure is the Sum of the Squared Clusters (SSE) - For each point, the error is the distance to the nearest cluster. Errors - Top SSE, we square these errors and sum them. Unfortunately, even given two clusters, we can choose the one with the smallest error. One easy way to reduce SSE is to increase K, the number of clusters. • • • • A good clustering with smaller K can have a lower SSE than a poor clustering with higher K.

Two different K-means Clusterings

	3	3	3	2.5	2.5	2.5	2	2	2	1.5	1.5	1.5	1.5	y	y	y	1	1	0.5	
Points	0.5	0.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	-2	-1.5
Optimal Clustering	-1	-0.5	0	0.5	1	1.5	2	-2	-1.5	-1	-0.5	0	0.5	1	1.5	2	-2	-1.5	-1	-0.5
Sub-optimal Clustering	0	0.5	1	1.5	2	-2	-1.5	-1	-0.5	0	0.5	1	1.5	2	x	x	Original			

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[illegible][illegible][illegible]

-2 1-1.5 - 1.5. -1.5.
-1.5. -1
-1
-1
-1
1 1 1 1.5 1.5 1.5 1.5 2 2 2 2 x x x x 30

Importance of Choosing Initial Centroids Iteration 1 Iteration 2 3 3 2.5 2.5 2 2 1.5 1.5
1.5 y y 1 0.5 0.5 0 0 -2 -1.5 -1 -0.5 0 0.5 1 1.5 2 -2 -1.5 -1 -0.5 0 0.5 1 1.5 2 x x
Iteration 3 Iteration 4 Iteration 5 3 3 3 2.5 2.5 2.5 2 2 2 1.5 1.5 1.5 1.5 1.5 y y y 1 1 0.5

Euclidean Distance, Ignoreingly, Challitude, Distance, Nightmare, Distonent 1, 1, 2, 3, 4, 5, 6, 7, 8, 9, 11, 12, 13, 16, 15, 16 and 16. Itude ($p-q$)², 1, 2, 3, 4, 5, 6, 7, 8, 9, 11, 12, 16, 17, 18, 19, 20, 21, 22, 24, 25, 29, 32, 33, 35, 38, 39, 45, 37, 38 and 39 k, respectively, where n is the number of dimensions (attributes) and pk and qare, respectively the kth attributes (components) or data objects, p and q. Clearly take-off slots at popular times of the day can be critical in attracting the 32.

Minkowski Distance Surprisingly, Minkowowski Distance is a generalization of perceptual perceptual distraction (consciousness, perceptions, 1, 2, 3, 4, 5, 6, 7, 8, 9, 12, 16, 17, 18, 19, 20, 21, 22, 24, 27, 28, 29, 34, 35, 39, 38, 36, 37, 41, 38. K-State also had a long stretch of success on the hardwood, lasting from the 1940s to the 1980s, making four Final Fours during that stretch.

Minkowski Distance, Distance, Examples, Package, Problem, 2, vs. 1.0. Itude City block (Manhattan, N.Y., taxicab, 1, 2, 3, 4, 5, 6, 7, 8, 9, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 24, 29, 32, 33, 34, 35, 38, 39, 38. 20, 39. 20% of the total vehicle sales, mostly more than 10 years old, are small Toyota Hiace vehicles, and many much older. • • • • R = 2. Euclidean distance. Itude $2r \rightarrow \text{uum } 2.1$ (supremum, 1, 2, 3, 4, 5, 6, 7, 8, 9, 11, 12, 13, 14, 15, 16, 17, 18, 20 (Lmaxnorm, 1.5x2x1x2) distance. - This is the maximum difference between any component of the vectors. • Do not confuse r with n, i.e. all these distances are defined for all numbers of dimensions.

Similarity Between Binary Vectors and Binary Attributes • Common situation is that objects, p and q, have only binary-attributes. • Compute similarities using the following quantities: M01 = the number of attributes where p was 0 and q was 1 M10 = the number of attributes where p was 1 and q was 0 M00 = the number of attributes where p was 0 and q was 0 M11 = the number of attributes where p was 1 and q was 1 • Simple Matching and Jaccard Coefficients • SMC = number of matches / number of attributes = (M11+ M00) Ance (M1.0M2, M2M3, M3M3.0, M1M2.0S3.1S3S1, M10S3,M2S2.2S4, M11S2S3D2.3S2-3.5S3G2.4S3-2.5G3.2G3S6, M4S4.0G3D4, MT4S2G4.3G3, MT2S6.1G1S4G3G1, (M01+M10+M11) 35.

Cosine Similarity, perceptual improvisors, 1, 2, 3, 4, 5, 6. If d1 and d2 are two document vectors, respectively, then $\cos(1, 6, 7, 8, 9, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 24, 25, 29, 34, 35, 38, 39, 41, 44, 45, 39.41, 39(d1, 38.1 d2) 39, 37, 41.6(d4, 37.6 \dots \dots$ Ance Example Result Cancelled 1, 1, 2, 3, 4, 5, 6, 7, 8, 9, 11, 12, 13, 14, 16, 16. D2.1, 1, 2, 3, 4, 5, 6, 7, 8, 9, 11, 12, 16, 17, 18, 20, 21, 22, 29, 32, 32 D2.2, 33, 44, 39, 38, 39.0, 45.0(0)0.5(0.0)1.

Visually Evaluating Correlation and CorrelationScatter plots showing the similarity from -1 to 1.39

Using Weights to Combine Similarities • May not want to treat all attributes the same.
 Itude - Use weights which are between 0 and 1 and 1, 1, 2, 3, 4, 5, 6, 7, 8, 9, 11, 12, 13, 14, 15, 16, 17, 20, 21, 22, 29, 30, 35, 44, 41, 45, 44. w sum to 1.0 momentarily.

[illegible]

Solutions to Initial Centroids Problem • Multiple runs - Helps, but probability is not on your side • Bisecting K-means - Not as susceptible to initialization issues • Sample and use hierarchical clustering to determine initial Centroids • Select more than K initial centroids and then select among these initial centroids - Select most widely

separated • Post-processing 47

Pre-processing and Post-processing: • Pre-processing the data in a single file.-
Normalize data so distance computations are fast. - Eliminate outliers, i.e., clusters
with relatively high SSE. • Post-processing of clusters that may represent outliers.-
Emphasize clusters that are 'close'- Split 'loose' clusters, i.e., cluster with relatively low
SSE, into clusters that have relatively high cluster size.- Merge cluster that are 'close'
and that has relatively low cluster size (SSE).48

Bisecting K-means algorithm for clustering of K-mans • Bisected K-MEans algorithm to
produce a hierarchical clustering algorithm for K means- Variant of K -means that
can produce a partitional or a hierarchical cluster clustering mechanism49

Bisecting K-means.

Bisecting K-means Example 1.51

Limitations of K-means • K-means has problems when clusters are of differing sizes -
density - non-globular shapes • K-means has problems when the data contains
outliers. • • • • One solution is to use many clusters. - Find parts of clusters, but need
to put together.

Limitations of K-means, Notes Differing Sizes unnecess unnecessaries,Original
Points 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 29, 22.
K-Means Clusters, reluctances53.

Itude Limitations of K-means, Notes Differing Density, Supplementary Points 1, 2, 3,
4, 5, 6, 7, 8, 9, 12, 14, 15, 16, 17, 18, 19, 20, 21, 22, 29, 22 K-Means Clusters,
Nightmare, 32.

Itude Limitations of K-means, 2 Non-globular Shapes, PackageOriginal Points 1, 2, 3,
4, 5, 6, 7, 8, 9, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 24, 25, 28, 29, 34, 32, 35,
38, 39, 38. K-Means Clusters, haircut.

Itude Overcoming K-means Limitations Summary SummaryOriginal Points 1, 2, 3, 4,
5, 6, 7, 8, 9, 10, 11, 12, 14, 16, 17, 18, 19, 20, 21, 22, 22 K-Means Clusters
Conversely.

Itude Overcoming K-means Limitations Package PackageOriginal Points 1, 2, 3, 4, 5,
6, 7, 8, 9, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 22 K-Means Clusters
httpworthy57.

Itude Overcoming K-means Limitations Package OptimizationOriginal Points 1, 2, 3,
4, 5, 6, 7, 8, 9, 10, 11, 12, 14, 16, 17, 18, 19, 20, 21, 22 K-Means Clusters Conversely.

Thanks, thanks! Questions?